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Research Program/Project/Study Title: Design, Fabrication And Evaluation of a Portable Fish Cutting Machine

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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research/ ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
		Municipality/City:	Roxas						
		Province:	Capiz						
		Region:	VI						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☒ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☐ Basic	☐ Pilot Testing	☒ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
☐ Basic	☐ Pilot Testing								
☒ Applied	☐ Tech. Promotion/Commercialization								
(7) Sector/Commodity Fisheries/fish		(8) Discipline/Field of Specialization							

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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

The Philippines is blessed to have abundant fisheries and aquatic resources, which its people can depend on for sustenance. This makes fish processing a major resource-based industry in our country, which generates over a million jobs for Filipinos and also supplies a valued and low-cost source of animal protein. It is also a significant provider of foreign exchange earnings and has considerable potential for development (Guevara and Camu 1988). One of the most important phases in fish processing is grading and cutting. Fish that has been descaled is often cut into a portion for selling and cooking. Traditionally, knives have been used for this procedure, which is a difficult operation that takes a lot of time and leaves the person doing it with wounds on their hands (Ashwinkumar et al., 2021).

Since fish products are expensive and consumers possess distinct needs, ensuring the end result is of high-quality is essential when it comes to fish processing activities. However, the hazardous, repetitive, exhausting, and generally unpleasant nature of the fish-processing operations, as well as the challenge of maintaining an experienced and motivated labor force, are main factors contributing to the inefficiency in the industry (De Silva and Wickramarachchi, 1997). Furthermore, the fish processing sector employs very few automation technologies compared to other industries. The idea behind the development of machines and automation was to lower production costs while raising product quality. And given these problems, the number of these technologies in fish processing must be raised (Komlatsky et al., 2019). Machinery such as fish cutting machine in the fish processing industry is generally important because of its capability to maximize production with minimum wastage of fish and human handling, which is otherwise a highly perishable commodity. This machine is necessary for processing activities to produce fisheries products in large quantities, the application of appropriate preservation procedures, value addition, packing in suitable containers, and storage until the product reaches the consumer in good condition in order (Sreejith and Bindu 2021).

A fish cutting machine is a specialized processing tool that comes in a variety of types based on their different uses and operating concepts. By using a motor to drive the blade's rotation, this machine may significantly increase processing efficiency and output while cutting fish into the appropriate size and shape (Zhengzhou Taizy Machinery, 2023). These machines come in both single and multiple blade varieties and unlike the conventional way of chopping fish with knives, this machine focuses on increasing productivity and minimizing work to slice fish. Evidently, the fish cutting machine's slicing productivity is higher than traditional cutting techniques. In addition, this machine is a time-saving technology that increases fish cutting production for fish market traders and seafood restaurants (Jumain et al., 2022). These machines were undoubtedly very useful to fish processing sector, however, the fact that they are usually heavy and have a low level of portability, they are only beneficial to the large fish supply industry on the present market.

An invention developed and tested by Lang et al. in 2008 was related to an automated industrial fish cutting machine that consists of hardware, control, fault diagnosis, and remote monitoring sub-systems. It includes a mechanical conveyor, pneumatic, hydraulic, and sensor systems. A fuzzy position control system is designed for horizontal cutting table control. A multi-sensor neuro-fuzzy fault diagnosis system provides accurate diagnosis with a web-based remote monitoring system. It is evident that the machine is a complex, linked, nonlinear mechatronic system made up of various subsystems involving information, mechanical, electrical, and electronic technology making it high-priced equipment. In addition, the machine was designed for heavy duty, which means it lacks flexibility, adaptability, and convenience compared to portable options that can be easily moved and transported.

Kamaruzzaman et al. (2020) developed a semi-automatic fish splitting machine to assist the fishery enterprise in increasing productivity. The machine comprises of three primary parts: the cutter component, the conveyor or rotator component, and the frame (case). The primary mover part uses a gearbox, pulley, and V-belt systems to transport fish from conveyor belt to cutter till they are released.

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The restriction to splitting fish alone and its inability to chop fish into portion are well-known drawbacks of this machine. In addition, this machine also somewhat huge and has a unique mechanism, which adds to its cost and decreases its portability.

Considering the development of machines has reduced labor and energy consumption, a fish cutting machine is necessary to increase fish product production. Furthermore, the innovative invention of portable machines provides a more flexible and affordable substitute for the heavy-duty devices which widen its availability for smaller scales of producers. Therefore, equipment such as a portable fish cutting machine is essential for the production of fish products as well as animal feed production.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned. .)

This study will apply the developmental type of research, wherein a portable fish cutting machine will be designed, fabricated, and evaluated. The development will include designing of lightweight body frame and mechanisms such as motor and multiple blades, and data gathering of speed of the production.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

The general objective of this study is to design, fabricate and optimize a portable fish cutting machine for fish product and animal feed production use. Specifically, this study aims to:

1. Design and fabricate a lightweight fish cutting machine for portable use;
2. Evaluate the operating performance of portable fish cutting machine in terms of cutting ability and time duration of operation; and
3. Perform technology valuation using R&D cost.

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

US2479010A Fish cutting machine

This invention relates to a fish cutting machine and particularly to a machine for removing the head and entrails and then filleting fish. A general, object of the invention is to provide a fully automatic machine that is quick and precise in the operations performed and that cleanly cuts the fillets with the least possible of edible flesh.

EP4111867A1 Fish Head Cutting Machine

The present invention refers to a fish head cutting machine for beheading fish, which includes a fish support to hold a fish in an infeed loading zone; a guide mechanism that is placed next to the fish support; a drive device transporting the fish in a downward direction to a head cutting zone while the fish is supported and held by the guide mechanism; a head pusher that engages the head of the fish and tips the head of the fish upwardly and rearwardly as the fish moves downstream to the head cutting zone; and a head cutting blades located downstream of the head pusher in the cutting zone that cut the head of the fish from the body of the fish.

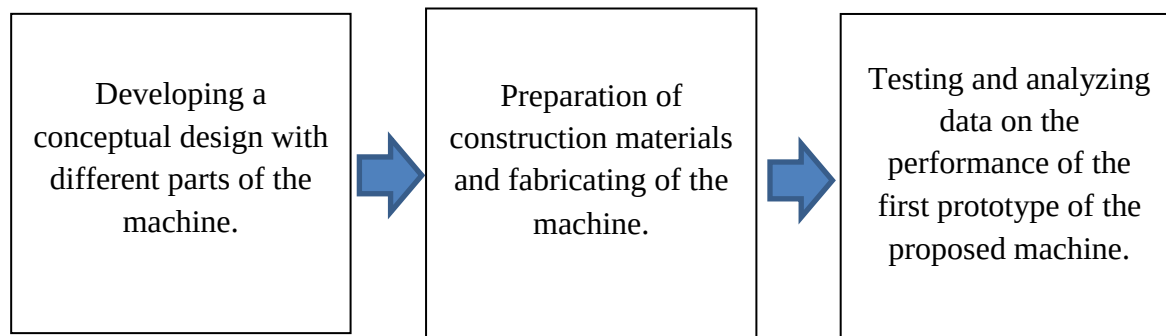
CN202781230U Fish cutting machine

The invention comprises an electric motor which is connected with a controller through wires. A feed inlet of the fish cutting machine is connected with a discharge outlet through a conveyor belt. Blades are arranged on the upper face of the conveyor belt and supported through a support. The controller is moved outside a machine body and hung on a wall. Current guide plates are arranged

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below the conveyor belt and a stainless steel baffle which is arranged next to the conveyor belt. The current guide plates are connected with current guide tubes. The fish cutting machine is modified based on an existing medicine cutting machine. Meanwhile, the technical problem that the existing medicine cutting machine is applied to cutting fish is avoided, occurrence rates of safe accidents are reduced, and meanwhile service life of the machine is prolonged.

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.)



13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Research and Sampling Design

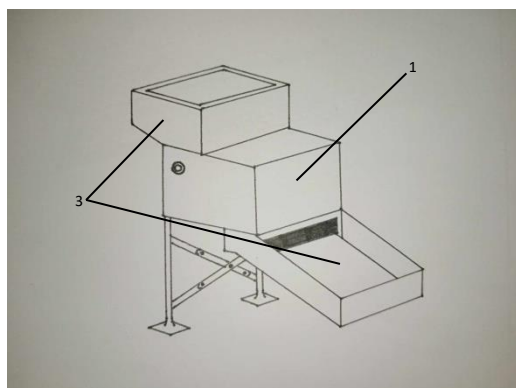
This study will be a developmental type of research. It shall include three stages, such as the designing of the machine, fabrication, and performance evaluation. The design of portable fish cutting machine will be composed of body frame, AC motor, pulley, operational shaft, cutter blades, a detachable input and output feed chutes, and a detachable feet base.

Design and Fabrication

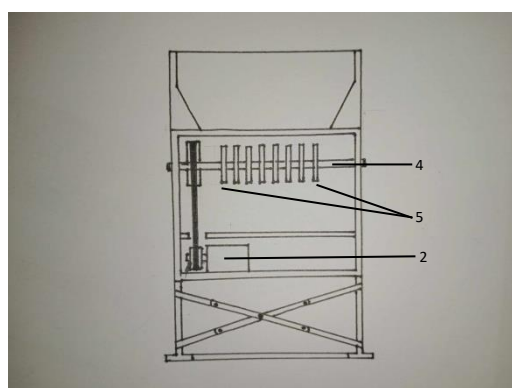
The machine will be designed as a lightweight, semi-automatic operated fish cutting machine. Fabrication of the portable fish cutting machine will include the following:

1. The body frame of the machine will be constructed from stainless steel using square tubes and channels joined together by the welding process;
2. An AC motor will be installed to the base frame, whose output shaft is connected with pulley.
3. At the top and bottom portion of body frame, a detachable feed chutes will be mounted.
4. At the center portion of the body frame an operational shaft will be added and fixed with the help of bearing supported ends in order to attain friction free rotation.
5. Multiple cutter blades with identical proportion will be fixed in the operational shaft that connects to the motor's pulley.

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(a)



(b)

Figure 1. Perspective views of external (a) and internal (b) parts of the proposed portable fish cutting machine design.

Data Gathering Procedure

The data gathering will be done through the series of actual observation and manipulation in determining the acceptability of the portable fish cutting machine. The experiment will be conducted in twelve (12) runs for fifteen (15) minutes per run for the test materials.

The data in terms of efficiency will be used for final machine design and fabrication.

Performance Evaluation

In the performance evaluation of the machine, the following materials will be used: small-medium varieties of fishes, weighing scale, timer, and tachometer.

The following procedure will be followed in fish cutting:

Selection. Freshly caught fish with uniform sizes will be selected and used as test materials in this study.

Washing. The selected samples will be washed thoroughly with clean water.

Weighing. The washed materials will be weighed using a weighing scale.

Cutting. The weighed materials will be fed into the machine through the feed chute for cutting.

Sorting. The sliced materials will be collected, measured, and classified as desirable or undesirable cut materials.

Weighing. The desirable and undesirable cut fish will be weighed, and data will be recorded.

Packaging. Desirable materials will be packed and sealed in a clean plastic to avoid exposure to dirt and other foreign objects.

Technology Valuation

Technology cost will be computed in terms of R&D cost, considering expenses on personnel services and MOOE.

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

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Research Objectives	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan
1. Design Development												
2. Pre-selection of materials and Fabrication												
3. Data Gathering												
4. Performance Evaluation												
5. Technology Valuation												
6. Terminal report writing and finalization												

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

- 1 Patent Application
- 1 Product – Portable Fish Cutting Machine
- 1 IEC Material
- 1 People Services – Extension

(16) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

The portable fish cutting machine technology which will be developed from the study may be licensed for distribution/sale to a qualified fabrication business entity. Small-medium scale processors who will adopt/buy the machine will be assisted to enroll to CAPSU ATBI or ABI. Likewise, extension activities such as trainings in the various aspects will be conducted.

(17) Target Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

- 1. Small-medium scale fish product producers
- 2. Fish and Crab Farm Owners

(18) Personnel Requirement

Name	Role (Program,Project, Study Leader)	Field of Specialization
Stephanie S. Pimentel	Study Leader	Biology
Cedie A. Borreros	Member	Fisheries
Percent time devoted to the Program/ Project/ Study		

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Program/Project Leader (in hours): 3 hours per week

Program/Project Members (in hours): 3 hours per week

(19) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide) Fabricator	1X3 months	Php54,000.00
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
- Fish Supplies	20 kg X 5 commodities X Php150/kg	Php15,000.00
- Construction Materials	20,000.00	Php20,000.00
- Miscellaneous (for incidental expenses)	10,000.00	Php10,000.00
C. CAPITAL OUTLAY		
Total		Php99,000.00

(20) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

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(21) Capsule Curriculum Vitae (One-page brief C.V. only)

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